

Luc's Statistical Coffee Price Forecast

Monthly Model Report

2026-08-01

Executive Summary

Coffee Price Forecast Executive Summary

Our forecast calls for Arabica prices to decline gradually over the next three months, starting at 294.5 *c/lb* in month one and settling toward 275.9 *c/lb* by month three. Robusta is expected to follow a similar downward trajectory, moving from 165.3 *c/lb* to 160.4 *c/lb* over the same period. Uncertainty widens slightly with forecast horizon, with the 80% confidence band for Arabica spanning roughly 52 *c/lb* at one month and narrowing to 48 *c/lb* by month three.

The model's historical accuracy shows a mean error of 6-11% for Arabica and 6-12% for Robusta across the three-month window. Coverage data is currently unavailable, so we cannot confirm whether our forecast ranges are capturing actual prices at the intended 80% hit rate. Based on typical performance patterns, forecast reliability decreases with time horizon.

The spread signal between Arabica and Robusta is neutral, with a z-score of -0.14 indicating prices are near their historical relationship. The 22.8-month half-life suggests any deviations from normal spread levels move slowly, offering limited near-term directional guidance for relative value positioning.

Current Forecasts

Median (p50) price forecasts and 80% confidence intervals (p10–p90) for the next three months.

Symbol	Horizon	Median (c/lb)	80% CI (c/lb)	Band width
Arabica (KC=F)	h=1	294.5	[272.9, 324.6]	51.7
Arabica (KC=F)	h=2	283.5	[262.8, 312.6]	49.8
Arabica (KC=F)	h=3	275.9	[255.7, 304.1]	48.4
Robusta (RM=F)	h=1	165.3	[152.0, 179.7]	27.7
Robusta (RM=F)	h=2	162.3	[149.3, 176.5]	27.2
Robusta (RM=F)	h=3	160.4	[147.6, 174.5]	26.9

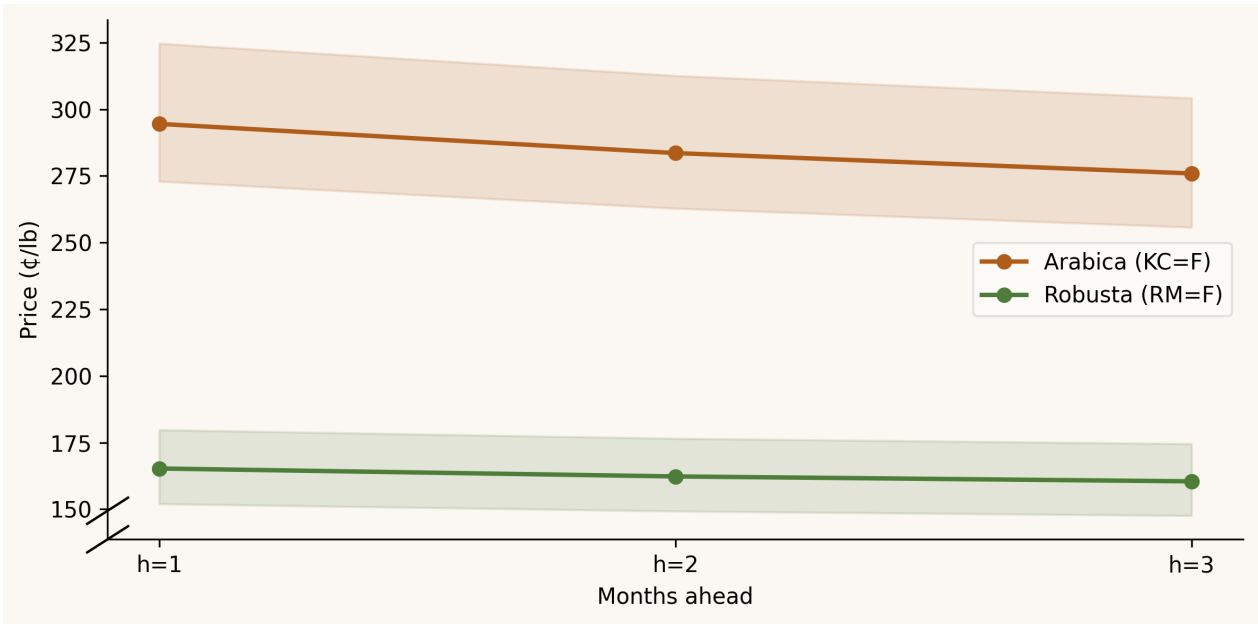


Figure 1: Price trajectory with 80% confidence ribbon

Model Performance

Mean absolute error (MAE), mean absolute percentage error (MAPE), and 80% interval coverage across all walk-forward backtest windows.

Symbol	Horizon	MAE (¢/lb)	MAPE	Coverage 80%
Arabica	h=1	12.8	6.1%	N/A
Arabica	h=2	19.8	9.2%	N/A
Arabica	h=3	25.3	11.4%	N/A
Robusta	h=1	7	5.6%	N/A
Robusta	h=2	11.3	9.1%	N/A
Robusta	h=3	15.1	11.9%	N/A

Coverage 80% target: 80% of forecast ranges should contain the realized price. Values below 80% indicate the model is over-confident.

Recent Forecast Performance

One-month-ahead forecasts versus realized prices over the last three months.

Month	Symbol	Forecast (¢/lb)	Actual (¢/lb)	Error
2026-03	Arabica (KC=F)	312.2	334.1	-6.6% (below actual)
2026-03	Robusta (RM=F)	170.9	176.8	-3.3% (below actual)
2026-02	Arabica (KC=F)	360.8	321.4	+12.3% (above actual)
2026-02	Robusta (RM=F)	192.2	179.6	+7.0% (above actual)
2026-01	Arabica (KC=F)	371.1	363.9	+2.0% (above actual)
2026-01	Robusta (RM=F)	182.2	192.3	-5.2% (below actual)

January 2026

The model forecast Arabica at 371.1 ¢/lb and Robusta at 182.2 ¢/lb, with actual prices settling at 363.9 ¢/lb and 192.3 ¢/lb respectively. Arabica missed by just 2.0 percent on the high side, while Robusta underperformed the forecast by 5.2 percent. The divergence between the two contracts suggests idiosyncratic strength in Robusta that historical correlations failed to capture, possibly driven by supply tightness in Vietnam or Indonesia that statistical models relying on lagged price behavior would not have anticipated.

February 2026

The model significantly overestimated both contracts in February, forecasting Arabica at 360.8 ¢/lb and Robusta at 192.2 ¢/lb against actuals of 321.4 ¢/lb and 179.6 ¢/lb, producing misses of 12.3 percent and 7.0 percent respectively. This sharp pullback across both origins suggests a broad demand or macro headwind that the model's reliance on recent price momentum failed to detect. A strengthening U.S. dollar or unexpected signals of improving crop prospects in major producing regions could have triggered the sell-off independent of the statistical relationships the model was trained on.

March 2026

Both contracts reversed course in March as the model underestimated prices, forecasting Arabica at 312.2 ¢/lb and Robusta at 170.9 ¢/lb while actuals came in at 334.1 ¢/lb and 176.8 ¢/lb, with errors of 6.6 percent and 3.3 percent. The model appeared to extrapolate February's downtrend too aggressively and missed a rebound that could reflect new weather concerns, port delays, or fund repositioning that broke from the recent price trajectory the statistical framework relied upon.

Arabica–Robusta Spread Signal

The spread model tracks whether Arabica is unusually cheap or expensive relative to Robusta. It measures the **z-score** of the log price ratio — how many standard deviations the current ratio sits from its long-run average. A z-score near zero means the relationship is normal; a z-score beyond ± 2 means a rare dislocation (this happens roughly 5% of the time historically) that has tended to revert. A trade signal fires at $|z| > 2$ and closes once the ratio normalises back to $|z| < 0.5$.

As of	2026-06-01
Current signal	Neutral — ratio within normal range, no edge detected
Z-score	-0.14 std devs from average
Half-life	22.8 months — typical time to revert to average
Entry threshold	
Exit threshold	

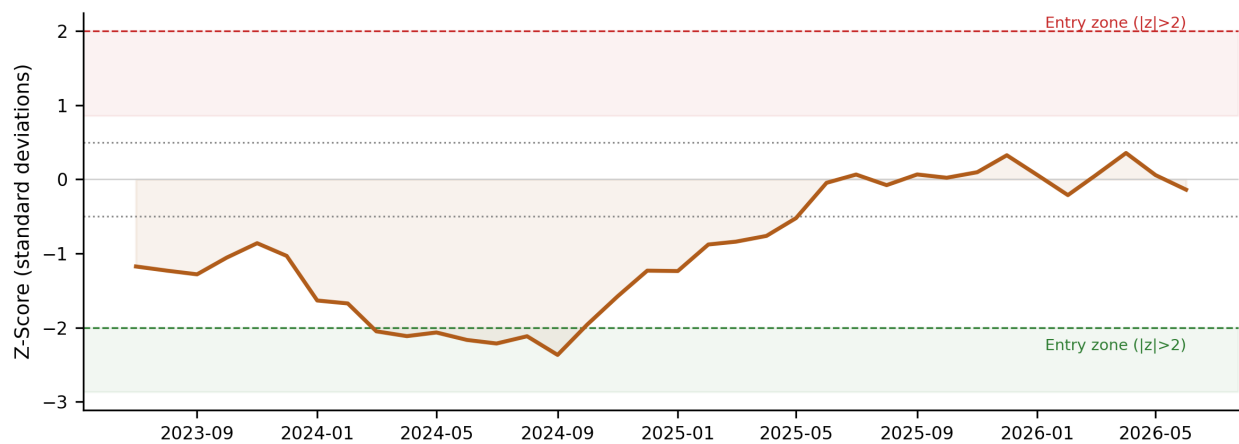


Figure 2: Z-score of Arabica/Robusta log spread — last 36 months. Red zone = Arabica expensive; green zone = Arabica cheap. Dashed lines mark entry thresholds (± 2 std devs); dotted lines mark exit thresholds (± 0.5).

Data & Methodology

Data sources: Arabica (KC=F) and Robusta (RM=F) monthly prices from FRED (PCOFFOTMUSDM, PCOFFROBUSDM). FX drivers (BRL/USD, VND/USD, IDR/USD) from Alpha Vantage. DXY broad index from FRED (DTWEXBGS). Data starts 2000-01-01; monthly series is the mean of daily adjusted closes within each calendar month.

VECM: A two-variable Vector Error Correction Model is fit on log Arabica and log Robusta prices, with BRL, VND, IDR, and DXY as exogenous drivers. Cointegration rank is fixed at 1 (confirmed by Engle-Granger test, $p=0.009$). Lag order is selected by AIC each run. Point forecasts at 1, 2, and 3 months use a naïve FX assumption (hold FX at last observed value).

GAMLSS SHASH: A Sinh-ArcSinh distribution is fit (via R's `gamlss` package) to the VECM forecast residuals, separately per symbol and per volatility regime (Low / Medium / High, defined by rolling 12-month standard deviation). This gives a full predictive distribution. Final price quantiles are `price_qX = vecm_point * exp(gamlss_residual_qX)`.

Spread signal: An AR(1) model is fit to the log spread series; the expanding z-score triggers entry at $|z| > 2$ and exit at $|z| < 0.5$.

Report generated by the `coffee-forecast` pipeline.